General Biology 1

First Semester

Supplementary Materials

Module 8

Repair, Regeneration, and Reproduction

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What I know

Directions: Read each situation carefully and choose the best answer.

1. Which process is primarily responsible for growth and repair of body tissues?

a. Meiosis
b. Binary fission
c. Mitosis
d. Fertilization

2. What is the main function of meiosis?

a. To produce identical body cells
b. To repair damaged tissues
c. To produce haploid gametes for reproduction
d. To increase cell size

3. Why is mitosis important in wound healing?

a. It reduces chromosome number
b. It produces genetically identical cells
c. It creates genetic variation
d. It stops cell division

4. Which statement best describes the importance of meiosis?

a. It maintains the same chromosome number in body cells
b. It produces identical daughter cells
c. It ensures genetic variation in offspring
d. It occurs only in damaged tissues

5. If mitosis did not occur in an organism, what would most likely happen?

a. The organism would reproduce faster
b. The organism would not grow or repair tissues
c. The organism would produce gametes
d. The organism would have more genetic variation

What I Need to Know

This supplementary module is designed to help you understand the how mitosis is essential for cell repair and regeneration and meiosis for reproduction. In this lesson, you will explore cell division is importance in maintaining life's homeostasis.

Lesson 8 Repair, Regeneration, and Reproduction

Living organisms rely on cell division to maintain life, and two key processes—mitosis and meiosis—play essential roles in repair, regeneration, and reproduction. Mitosis is responsible for producing genetically identical cells, making it vital for the growth of organisms, the repair of damaged tissue. On the other hand, meiosis is involved in reproduction. Together, mitosis ensures the maintenance and survival of the individual organism, while meiosis supports the continuation of the species through reproduction.

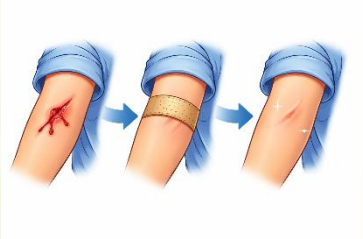
After going through this supplementary module, you are expected to:

1. describe the role of mitosis in organismal growth, tissue repair, and regeneration of damaged cells;
2. explain how meiosis contributes to sexual reproduction by producing haploid gametes; and
3. analyze the importance of mitosis and meiosis in maintaining life processes and ensuring genetic continuity and variation.

What's In

Activity 1: "From Damage to Recovery: The Mitosis Effect"

Directions: For each situation, identify how mitosis is involved by describing its role in the process

Situation	Role of Mitosis	Outcome
		

Guide Questions:

1. How does mitosis help in each situation (growth, wound healing, and tissue regeneration)? _____
_____
_____
2. What happens to the cells before, during, and after mitosis in these processes? _____
_____
_____
3. Why is it important that the new cells produced by mitosis are identical to the original cells? _____
_____

What's New

Activity 2: "The Chromosome Shuffle: Creating Life Through Meiosis"

Directions: Read carefully the scenario given and accomplish the task intended to complete this activity.

Scenario

You are a genetic scientist studying how organisms reproduce. A couple is curious about how their child inherits traits from both parents. You must explain how meiosis produces haploid gametes (sperm and egg cells) and how this process ensures genetic variation in their future child.

Task

You will simulate the process of meiosis using colored paper strips or chromosomes models. You will demonstrate how one diploid cell undergoes two divisions to form four haploid gametes, showing key stages such as crossing over and independent assortment.

Guide Questions:

1. What happens to the chromosome number during meiosis?

   _____

2. How are haploid gametes formed from a diploid cell?

   _____

3. Why is it important that gametes are haploid in sexual reproduction?

   _____

4. How does meiosis contribute to genetic variation?

   _____

Put your colored paper strips/chromosome model here

What's More

Activity 3: "Cell Division Showdown: Growth vs Reproduction"

Directions: Complete the table below about the importance of mitosis and meiosis in maintaining life processes

Function	Mitosis	Meiosis
Growth		
Repair		
Regeneration		
Reproduction		

Increases the number of body cells for growth and development

Does not function in repair

Does not contribute to growth

Produces new cells to restore lost or damaged tissues

Replaces damaged or dead cells with identical cells

Does not function in regeneration

Produces haploid gametes for sexual reproduction

Allows regrowth of tissues or body parts

Guide Questions:

1. How does mitosis support growth, repair, and regeneration in organisms? _____
_____
_____
2. Why is meiosis essential for reproduction but not for growth or repair?
_____
_____
_____
3. How do the roles of mitosis and meiosis complement each other in maintaining life? _____
_____
_____

To experience **MORE** interactivity in this module, scan this QR code.

What I've Learned

Directions: Encircle the letter of the correct answer.

1. A patient has a wound that does not heal properly because cell division is impaired. Which process is most likely affected, and why?
   a. Meiosis, because gametes are not formed
   b. Mitosis, because new identical cells are not produced
   c. Binary fission, because bacteria are not dividing
   d. Fertilization, because chromosomes are not combining
2. Why would replacing meiosis with mitosis in reproductive organs be problematic?

   a. It would produce haploid gametes
   b. It would reduce the chromosome number too much
   c. It would produce diploid gametes, causing chromosome doubling after fertilization
   d. It would stop cell division completely

3. A species loses the ability to undergo mitosis but can still perform meiosis. What is the most likely consequence?

   a. The organism can grow but not reproduce
   b. The organism can reproduce but cannot grow or repair tissues
   c. The organism cannot reproduce or grow
   d. The organism will have more genetic variation

4. Which situation best demonstrates the complementary roles of mitosis and meiosis?

   a. Mitosis produces gametes, while meiosis repairs tissues
   b. Mitosis repairs tissues, while meiosis produces gametes for reproduction
   c. Both processes produce identical cells for growth
   d. Both processes reduce chromosome number

5. If meiosis did not introduce genetic variation, what would be the most likely long-term effect on a species?

   a. Increased survival and adaptation
   b. No change in reproduction
   c. Reduced ability to adapt to environmental changes
   d. Faster growth of organisms